

Nifty100 Data Foundation

Analyst Guide

A practical guide to the Streamlit dashboard, PDF tearsheets, and REST API built on top of the Nifty 100 fundamentals data foundation. Covers 92 companies across 10 broad sectors, 2019-2024.

Sprint 6 - Day 44 Documentation Deliverable

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1. Overview & Getting Started

The Nifty100 Data Foundation project loads fundamentals for 92 Nifty 100-index companies (2019-2024) into a local SQLite database (data/nifty100.db), then layers three analyst-facing tools on top of it:

- **Streamlit dashboard** (port 8501) - 8 interactive screens for browsing, screening, and comparing companies visually.
- **FastAPI REST API** (port 8000) - 16+ JSON endpoints for programmatic access, integrations, or scripting.
- **PDF tearsheets** - one-page-per-company summary reports, pre-generated for all 92 companies under reports/tearsheets/.

Starting the dashboard

```
streamlit run src/dashboard/app.py
```

Opens automatically in your browser at <http://localhost:8501>.

Starting the API

```
uvicorn src.api.main:app --port 8000
```

Interactive API docs (Swagger UI) are then available at <http://localhost:8000/docs>, and the raw OpenAPI schema at [docs/openapi.json](http://localhost:8000/docs/openapi.json).

Both processes read the same SQLite file directly and don't talk to each other over HTTP, so they can be started, stopped, or restarted independently without port conflicts.

2. Using the Streamlit Screener

The Screener screen (pages/03_screener.py) is the fastest way to go from '92 companies' to 'the handful I actually want to look at'.

The 10 sliders

- ROE min - minimum return on equity (%)
- D/E max - maximum debt-to-equity ratio
- FCF min - minimum free cash flow (Cr)
- Revenue CAGR min - minimum 5-year revenue CAGR (%)
- PAT CAGR min - minimum 5-year profit CAGR (%)
- OPM min - minimum operating profit margin (%)
- P/E max - maximum price-to-earnings ratio
- P/B max - maximum price-to-book ratio

- Dividend Yield min - minimum dividend yield (%)
- ICR min - minimum interest coverage ratio

The 6 presets

Click a preset button to auto-fill every slider at once, then fine-tune from there if you want:

- **Quality** - high ROE, low leverage, consistent margins.
- **Value** - low P/E and P/B relative to fundamentals.
- **Growth** - high revenue and PAT CAGR.
- **Dividend** - high, sustainable dividend yield.
- **Debt-Free** - zero or near-zero debt-to-equity.
- **Turnaround** - improving trend on a previously weak metric (e.g. declining D/E year-over-year).

The results table updates live as you move any slider, shows a 'N companies match your filters' count above it, and every visible column can be exported with the **CSV download** button - useful for taking a shortlist into Excel or another tool.

3. Navigating Each Dashboard Screen

3.1 Home

Landing page. Shows 6 summary KPI tiles (average ROE, median P/E, median D/E, total companies, median 5yr revenue CAGR, debt-free company count), a sector breakdown donut chart, and a top-5 companies by composite quality score table. Use the year selector in the sidebar (2019-2024) to see how these headline numbers looked in past years.

3.2 Company Profile

Search any company by name or ticker (autocomplete as you type). Shows a company card (sector, sub-sector, NSE ticker, description), 6 KPI tiles, a 10-year revenue/net-profit bar chart, a ROE/ROCE dual-axis line chart, and pros/cons as green-check / red-cross badges. An unknown ticker shows a friendly 'Ticker not found - please try another' message rather than an error.

3.3 Screener

See Section 2 above for full detail on sliders and presets.

3.4 Peer Comparison

Pick one of the 11 peer groups from the dropdown to see a radar chart (the selected company's 8 metrics vs. its peer group average) and a side-by-side KPI table for every company in that group, with the designated benchmark company's row highlighted.

3.5 Trend Analysis

Search a company, then choose up to 3 metrics to overlay on a 10-year line chart, with YoY % change annotated at each data point - good for spotting inflection points (e.g. margin compression, a debt paydown, a growth re-acceleration).

3.6 Sector Analysis

Pick a sector to see a bubble chart (X = revenue, Y = ROE, bubble size = market cap, colour = sub-sector) plus a sector median KPI bar chart underneath - a quick way to spot outliers within a sector.

3.7 Capital Allocation Map

A treemap of all 92 companies grouped into 8 capital allocation patterns (e.g. reinvestment-heavy, dividend-heavy, debt-reduction-focused). Click a pattern to see the list of companies in it.

3.8 Annual Reports

Search a company to see every year it has an annual report on file, with a clickable BSE PDF link. If a link is broken (returns a 404), it's shown with a red 'Report unavailable' badge instead of a dead link.

4. Generating PDF Tearsheets

Every company has a pre-generated, one-page tearsheet PDF under `reports/tearsheets/{TICKER}_tearsheet.pdf` (92 files, each 30 KB+). To regenerate all of them from current data:

```
PYTHONPATH=. python3 -c "  
from src.reports.tearsheet import batch_generate_tearsheets  
batch_generate_tearsheets('data/nifty100.db')  
"
```

Each tearsheet includes revenue/profit history, ROE/ROCE trend, a balance-sheet snapshot, a cash-flow waterfall, and the same pros/cons shown on the Company Profile dashboard screen. You can also download a single company's tearsheet directly from the dashboard's Company Profile screen, or via the API (see below).

Companies that were skipped during a batch run (missing minimum data) are logged to `output/skipped_tearsheets.csv` with a reason column, so nothing fails silently.

5. Calling the REST API

All endpoints are namespaced under `/api/v1`. Every route returns JSON. Full interactive docs live at <http://localhost:8000/docs> once the server is running.

Health check

```
curl http://localhost:8000/api/v1/health
```

List all companies

```
curl http://localhost:8000/api/v1/companies
```

Search companies by name

```
curl "http://localhost:8000/api/v1/companies?search=Tata"
```

Full company profile

```
curl http://localhost:8000/api/v1/companies/TCS
```

P&L; history for a date range

```
curl "http://localhost:8000/api/v1/companies/TCS/pl?from_year=2020&to_year=2023"
```

Download a tearsheet PDF

```
curl -o TCS_tearsheet.pdf http://localhost:8000/api/v1/companies/TCS/tearsheet
```

Run the ad-hoc screener

```
curl "http://localhost:8000/api/v1/screener?min_roe=15&max_de=1&max_pe=40"
```

Run a named preset

```
curl "http://localhost:8000/api/v1/screener/run?preset=Quality"
```

Sector medians

```
curl http://localhost:8000/api/v1/sectors
```

Companies in a sector

```
curl "http://localhost:8000/api/v1/sectors/Information%20Technology/companies"
```

Peer group detail with percentile ranks

```
curl "http://localhost:8000/api/v1/peers/IT%20Services"
```

Peer radar comparison for a company

```
curl http://localhost:8000/api/v1/companies/TCS/peers/compare
```

Historical valuation multiples

```
curl http://localhost:8000/api/v1/market-cap/TCS
```

Portfolio-wide percentile stats

```
curl http://localhost:8000/api/v1/portfolio/stats
```

Annual report links for a company

```
curl http://localhost:8000/api/v1/companies/TCS/documents
```

A ready-to-import Postman collection covering every GET endpoint (with example values pre-filled) is at [docs/postman_collection.json](#).

6. Troubleshooting Common Issues

Symptom	Likely cause	Fix
Dashboard shows 'Ticker not found'	Typo, or the ticker uses a different symbol than expected (e.g. <code>WVE</code> vs <code>WVO</code>)	Use the ticker symbol as shown on the Company Profile screen instead
API returns 404 for a company	<code>company_id</code> is case-sensitive and must match exactly (e.g. <code>Apple</code> vs <code>apple</code>)	Copy/paste the company name exactly to find the exact <code>company_id</code>
Screener returns 0 matches	Sliders/filters are set too strictly for the current data	Loosen one filter at a time, or start from a preset and adjust from there
Chart overflows the page width	Browser window narrower than the chart's fixed width	All dashboard charts use <code>use_container_width=True</code> ; if one doesn't, tell us
'Address already in use' starting <code>Streamlit</code>	A previous instance is still running on that port	Find and stop it (ls of -i :8501 or :8000), or start on a different port with <code>streamlit run . --server.port 8000</code>
Tearsheet download 404s	Tearsheet PDF hasn't been generated for that company yet	Run <code>company_batch_generate_tearsheets()</code> (Section 4) and check output/sk
CSV export from the screener looks garbled	Excel reads UTF-8 without a BOM for non-ASCII characters	Open the CSV via Excel's Data > From Text/CSV import instead of d
pytest fails with 'ModuleNotFoundError'	Test is imported from <code>src.*</code> but the project root isn't on <code>PYTHONPATH</code>	Run <code>PYTHONPATH=. python3 -m pytest tests/</code> from the project root

For anything not covered here, check `output/perf_notes.md` for known performance characteristics, or `docs/acceptance_checklist.pdf` for the full list of verified acceptance gates and their current status.

7. Glossary of Metrics

Quick reference for every KPI abbreviation used across the dashboard, tearsheets, and API responses.

Term	Meaning
ROE	Return on Equity - net profit / shareholder equity, as a %.
ROCE	Return on Capital Employed - a broader profitability measure than ROE that also accounts for debt.
ROA	Return on Assets - net profit / total assets, as a %.
D/E	Debt-to-Equity ratio - total debt / shareholder equity. 0 means debt-free.
OPM	Operating Profit Margin - operating profit / revenue, as a %.
FCF	Free Cash Flow - cash from operations minus capital expenditure, in Rs. crore.
FCF Yield	FCF / market capitalisation, as a % - how much free cash a company generates relative to its price.
CAGR	Compound Annual Growth Rate - the smoothed annual growth rate over a multi-year period (e.g. Revenue CAGR).
ICR	Interest Coverage Ratio - operating profit / interest expense. Undefined (shown as N/A) when a company has no interest expense.
P/E	Price-to-Earnings ratio - share price / earnings per share.
P/B	Price-to-Book ratio - share price / book value per share.
EV/EBITDA	Enterprise Value / EBITDA - a valuation multiple that, unlike P/E, is capital-structure neutral (accounts for debt).
Composite Quality Score	A single 0-100 score blending ROE, D/E, FCF, growth, and margin metrics, winsorized within each metric to 0-100.
Overvaluation Flag	Caution if P/E > 1.5x the company's sector median P/E; Discount if P/E < 0.7x the sector median; otherwise None.
Capital Allocation Pattern	One of 8 archetypes (e.g. reinvestment-heavy, dividend-heavy) describing how a company has historically deployed capital.

8. Appendix: Database Tables

For analysts who want to query `data/nifty100.db` directly (e.g. with a SQLite browser, or pandas' `pd.read_sql`), here's what each table holds:

Table	Contents
companies	One row per company - name, ticker, description, latest ROE/ROCE snapshot.
sectors	broad_sector and sub_sector classification per company.
profitandloss	Yearly P&L line items (2019-2024) per company.
balancesheet	Yearly balance sheet line items per company.
cashflow	Yearly cash flow statement line items per company.
financial_ratios	Pre-computed KPIs per company per year (ROE, D/E, OPM, etc.).
market_cap	Yearly market cap and valuation multiples (P/E, P/B, EV/EBITDA).
stock_prices	Daily/periodic stock price history.
peer_groups	Peer group membership (11 groups) and benchmark-company flag.
peer_percentiles	Percentile rank per company, per peer group, per metric (10 metrics).
prosandcons	Auto-generated pros/cons text per company, with a confidence score.
documents	Annual report links per company per year.

See `db/schema.sql` for full column-level definitions, primary/foreign keys, and indexes.